

River Deep Mountain AI

Model Benchmarking Report: Flow (BB1A)

Model Benchmarking Report - Flow (BB1A)

FINAL REPORT

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Glossary

BFI	Base Flow Index
CERF	Continuous Estimation of River Flows
FDC	Flow Duration Curve
LSTM	Long Short-Term Memory
NRFA	National River Flow Archive
NSE	Nash-Sutcliffe Efficiency
PBIAS	Percent Bias
RDMAI	River Deep Mountain AI
RMSE	Root Means Square Error
TSEP	Time Series of Exceedance Probabilities
UKCEH	UK Centre for Ecology and Hydrology
WHS	Wallingford HydroSolutions



Summary

Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
Model Approach	QUBE flows are generated by the deterministic CERF2-HadUK daily rainfall-runoff model. This model generates ~11,000 time series of exceedance probabilities (TSEP) catchments across the United Kingdom and Ireland. Then, for any ungauged catchment, QUBE produces a natural flow duration curve (FDC) based on catchment characteristics. The timeseries flow is generated by stepping through the geographically closest TSEP and selecting the corresponding flow rate from the FDC.	The RDMAI Open Flow Model is a machine learning model. It relies primarily on the CAMELS-GB dataset containing hydro-meteorological timeseries and landscape attributes for 671 catchments in Great Britain. Time series attributes are handled by a Long-Short-Term Memory (LSTM) layer, whereas static attributes are handled by an encoder. The LSTM layer works according to a 30-day sliding window to consider antecedent effects on flow. For each 30-day sliding window, the outputs of the LSTM layer and encoder are input into the model, and a single flow prediction for each catchment is produced, corresponding to the last day in the 30-day sliding window.
Applications	Generally available QUBE flows are naturalised flows (access to human influenced flows is restricted to the regulator due to confidentiality of data used). Naturalised flows are most useful for input into water resources models, used for water resources management modelling, such as infrastructure development planning. Another application of QUBE flows can be estimating pollutant dilution for combined sewer overflow (CSO) filtering – a first step in deciding whether a CSO requires more in-depth analysis due to its impact on instream water quality.	The RDMAI Open Flow Model provides estimates of human impacted flows. The RDMAI Open Flow Model estimations could feed into SAGIS as an alternative to CEH naturalised flows. Currently, flow estimates in SAGIS are naturalised flows provided by CEH. SAGIS then adds human influences, and flows can be calibrated by comparison to gauged flows. If the RDMAI Open Flow model was used, this could avoid the need for further calibration in SAGIS since the RDMAI estimations already consider human influences. Furthermore, flow estimations, given that they account for human interventions, could be used to estimate the likely impact of interventions on ecology, fisheries, and/or protected areas and could have future application for assessing the potential impact of interventions for WFD. The RDMAI Open Flow Model flow estimations



Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
		could also be used for predicting floods or for sustainable water resources management. The flow estimates could also be input into water quality models to represent flow dilution and pollution residence time. There is potential for bringing the flows of the RDMAI Open Flow Model into the RDMAI Open Risk Map as well.
Geographic Constraints	QUBE flows are available for the United Kingdom and Ireland.	Whilst the model inputs come from the CAMELS-GB dataset , which covers Great Britain, the model was only trained using gauged flows from the EA Hydrological Data Explorer . Consequently, the RDMAI Open Flow Model can only estimate flows for England.
Temporal Resolution	The QUBE model provides daily average flow rate in m ³ /s.	The RDMAI Open Flow Model provides daily average flow rate in m ³ /s.
Data Inputs Required	QUBE flows can be accessed from a website . A paid-for licence is required to access QUBE flows. Since the user does not need to run the model, no data inputs need be considered.	The RDMAI Open Flow Model requires the CAMELS-GB dataset or datasets with equivalent data to that as the CAMELS-GB as input.
Model Performance	QUBE flows were generally a good match to gauged flows for the benchmark catchments. Instances where QUBE flows over-estimated gauged flows, could be attributed to the impacts of reservoirs in the contributing catchment (remembering that the QUBE flows assessed are naturalised flows). The NSE ranged from -0.46 to 0.96 and PBIAS ranged from -41.01% to 10.44%.	For the six benchmark catchments assessed, the RDMAI Open Flow Model provided reasonable estimates of gauged flows (representing the temporal pattern in gauged flows) for three catchments and good estimates (representing both temporal patterns and frequencies of magnitudes in gauged flows) for two catchments. NSE ranged from -0.73 to 0.89 and PBIAS ranged from -130.55% to 93.63%. The RDMAI Open Flow Model relies primarily on the CAMELS-GB dataset to inform estimated river flow. The benchmarking results indicate the need to further study and cross-check the CAMELS-GB dataset with other information to determine if there are any



Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
		inconsistencies or missing information such as the presence of reservoirs and reservoir capacity. In this way, additional input datasets to the RDMAI Open Flow Model could be explored in the future, to try and increase the accuracy of the model.
Cost & Time	Extracting QUBE flows take about 30 minutes per location. Therefore, extracting flows for 25 sites would take around 12.5 hrs. It should be noted that at present, QUBE is only available for lease on a regional basis, aimed at use by the UK regulators and water companies. These leases may be negotiated on an individual basis. Since WHS does not openly advertise the costs of QUBE, it is not possible to provide a cost estimate of QUBE in this report. An alternative model that can provide estimates of human impacted flow is the Water Evaluation and Planning (WEAP) model. WEAP includes built-in rainfall-runoff hydrology models, and can also represent human impacts, such as abstraction. A multi-user corporate licence for WEAP costs \$10,000/year.	Installing the RDMAI Open Flow Model and its dependencies and running through a sequence of scripts to download and prepare data, and run the trained model for a prepared example, takes 1 -2 hours. Most of this time is for preparing the datasets applicable to the entire Great Britain – running the trained model takes seconds. Training the model would require the steps mentioned above plus the time taken to train the model and was not assessed in this report. The RDMAI Open Flow Model is Open Source, free and uses open datasets. Therefore, there are no licence costs associated with running the model
Strengths	QUBE is recognised by the regulators of England, Wales, Scotland, Northern Ireland and Ireland, as well as by water companies in the UK, as the best-practice tool for estimating historical flows – both the long-term flow statistics and daily average flow timeseries. QUBE can also generate climate change-adjusted flow estimates based on the UKCP18 eFLaG meteorological datasets for England, Scotland and Wales.	The RDMAI Open Flow Model is open source, free and uses openly available input data. In addition, estimated flows consider human influences on flow rate, including abstractions and discharges. Therefore, the model simulated flows have the potential to reflect the influences of human impacts such as reservoirs. Consequently, flow estimations from the RDMAI Open Flow Model could be fed into SAGIS as an alternative to naturalised flows provided by CEH, thereby avoiding the need for calibration. The estimated flows also have potential application in flood modelling, wastewater planning, water resources planning, and for assessing the potential impacts of



Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
		interventions on ecosystems and for WFD. The scripts provided with the model also allow various graphing and mapping output. It is a known problem that a lot of EA monitoring for water quality is not co-located with flow measurements. Water quality is explicitly linked to water quantity, as flow regulates water quality processes, including pollution pathways, dilution and residence time. The ability to generate flow estimates at water quality monitoring points, using the RDMAI Open Flow Model, can help increase our understanding of the relationships between water quantity and quality, such as the possible sources of pollution, and can help set water quantity and quality boundary conditions for water quality modelling.
Limitations	Apart from to the regulator, available QUBE flows are naturalised flows. Human impacts to flows can be substantial in some catchments, considering reservoirs, transfers, abstractions and discharges, which naturalised flows will not reflect. Currently, QUBE flows are only available between 1955 and 2021. QUBE flows are a paid-for product, requiring a licence.	The RDMAI Open Flow Model gives reasonable estimates of gauged flows for most catchments - around 80% of the 123 catchments assessed in the validation. However, model shortcomings in predicting flows for the remaining 20% of catchments must be recognised. The reasons for the inability of the current RDMAI Open Flow model to adequately represent flows in this minority of catchments still need to be determined.



1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Flow Model Overview

Data for river catchment flow rate are essential for managing both river water quantity and water quality. From a quantity perspective, flow rate data are essential for predicting floods and for sustainable water resources management. From a water quality perspective, flow regulates catchment pollutant pathways to surface water, instream dilution of pollutants, and pollutant residence times in surface water. However, gauged river flows can be sparse or missing in many catchments, particularly for smaller or remote rivers. Additionally, the implementation of river flow monitoring programs is expensive. For example, Thriving Catchments suggest that detailed flow monitoring for two catchments (~26 flow gauges in total) costs upwards of £300,000 annually.

Model simulated flows act as an important surrogate for gauged data. Flow models can also allow future scenario analysis, which is of particular relevance given concerns over a changing climate. While deterministic hydrological modelling is a mature field, many existing models are limited in scalability and timeliness.

The RDMAI Open Flow Model aims to estimate daily average flow rate in ungauged catchments at a higher accuracy than those provided by current surrogate practices. The model uses a machine learning approach, leveraging both static and dynamic data sources.

The RDMAI Open Flow Model is primarily trained on catchment attributes contained in the [CAMELS-GB dataset](#). This dataset contains hydro-meteorological timeseries and landscape attributes for 671 catchments in Great Britain. These catchments are delineated as land surface areas with outlets at a subset of [National River Flow Archive \(NRFA\)](#) flow gauges. Therefore, these two datasets provide the model static/dynamic catchment inputs and observed flow data required, respectively, for model training. Given the unique hydrological characteristics of chalk streams, the RDMAI Open Flow Model also considers catchments defined as chalk streams in the model training.

The model architecture is composed of two branches based on the broad characteristics of the data in the CAMELS-GB dataset:

- Static catchment attributes, like land use, are handled by an encoder.
- Dynamic - Timeseries catchment attributes, like weather, are processed using a Long Short-Term Memory (LSTM) layer.

In the LSTM layer, the model has a sliding 30-day window to consider antecedent influences affecting flow. For each 30-day window, the outputs of both branches are concatenated, and



the model produces a single flow prediction for each catchment, corresponding to the last day in the 30-day window.

Model validation showed that the model gives relatively accurate predictions of measured average daily flow for most catchments [80% of catchments; mean Nash Sutcliffe Efficiency (NSE) of 0.54 to 0.62], while flow tends to be under-predicted for the remaining catchments. The poorly performing catchments also seem to be dispersed evenly across England, with no clear geographic pattern. Due to limitations in obtaining Qube licences for catchments, our benchmarking catchments do not allow the relative performances of the models across different catchment typologies to be assessed.

More information on the RDMAI Open Flow Model is available on the [RDMAI GitHub](#) page.

1.1.2 Overview of Existing Industry Model(s)/Tool(s)

QUBE

QUBE is recognised by the regulators of England, Wales, Scotland, Northern Ireland and Ireland, as well as by water companies in the UK, as the best-practice tool for estimating flows. The tool allows estimated river flows for any river reach in the United Kingdom and Ireland. QUBE is a licenced product owned by Wallingford HydroSolutions (WHS). Licences for using QUBE are provided on a regional basis, recognising that water companies have operational areas.

QUBE flows are accessible from a web-based system, which allows searching by place name, catchment, feature names and grid reference.

QUBE produces a natural Flow Duration Curve (FDC) for a catchment based on catchment characteristics, such as runoff and soil properties. The generalised Continuous Estimation of River Flows (CERF) (EA, 2008) rainfall-runoff model has been used to produce daily timeseries of flow percentiles for ~11,000 catchments in Great Britain (GB). These are called TSEP (time series of exceedance probabilities) catchments. Within QUBE, choosing the most appropriate TSEP is based on distance from the ungauged catchment and similarity in catchment characteristics. Once the most appropriate TSEP is chosen, the time series of flow is generated by stepping through the time series of flow percentiles and selecting the corresponding flow rate from the FDC.

In recent years, the UK Centre for Ecology and Hydrology (UKCEH) has produced gridded potential evaporation datasets based on the gridded HadUK observed meteorological datasets produced by the Met Office. The CERF rainfall-runoff model was redeveloped using the HadUK precipitation dataset and UKCEH potential evaporation dataset, as well as gauged flow data from 472 flow gauges, with the resulting model referred to as the CERF2-HadUK (WHS, 2024) daily rainfall-runoff model.

Flows generated by the QUBE model have been chosen for benchmarking, given the best-practice status of this tool amongst regulators and water companies in the UK. The CERF2-



HadUK daily rainfall-runoff model has been shown to have good performance when compared to observed flows for 472 gauges across the UK. A limitation of the current version of QUBE is that it provides historical flows between 1955 and 2021.

More information on QUBE can be found on the [QUBE online portal](#).

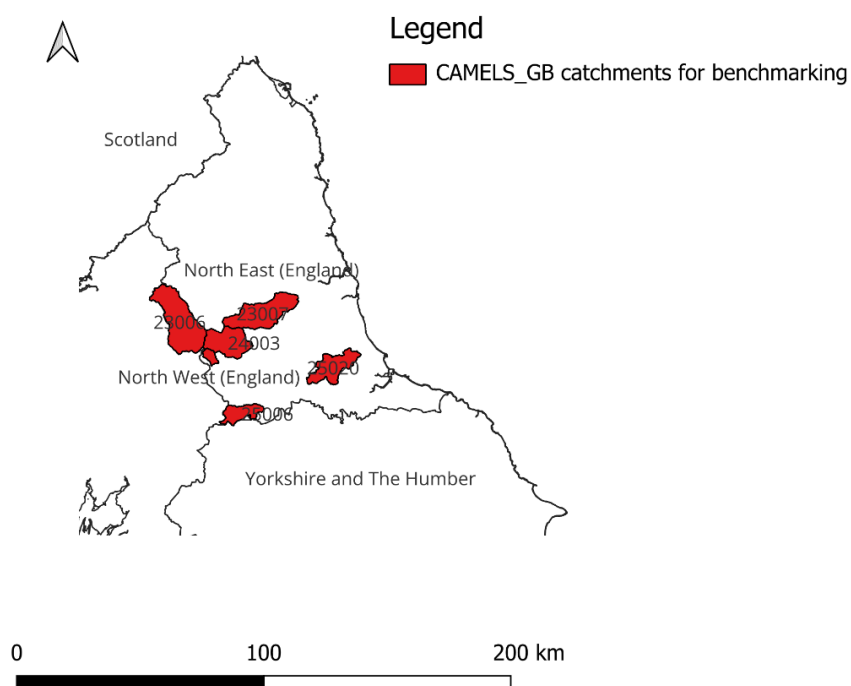
1.2 Catchment Selection

It should be noted that catchments available for benchmarking were limited to a subset of the catchments used for model validation. This is because benchmarking should be using catchments that were not part of the RDMAI Open Flow Model training dataset – ensuring that the RDMAI Open Flow Model had not ‘seen’ the benchmarking catchments before.

The RDMAI Open Flow model estimates of flow were benchmarked against QUBE estimates of flow. QUBE is a licenced product, requiring payment for extracted flows.

The benchmarking originally aimed to use a representative sample of catchments from across England, so that national variability in catchment characteristics (e.g. rainfall, base flow index (BFI), built-up area, etc.) could be accounted for. However, the benchmarking exercise only had access to Northumbrian Water Limited’s QUBE licence, which limits extraction of QUBE flows to the operational area of Northumbrian Water Limited (essentially, North-East England). Consequently, only a limited number of benchmarking catchments were available. The positions of the catchments in North-East England are shown in Figure 1.1, while Table 1.1 provides a summary of the main characteristics of these catchments. Note, there is no open dataset of BFI available for Great Britain. For this report, the BFI estimates given in the open NRFA gauge metadata information were averaged by operational catchment.

Figure 1.1 Position of benchmarking catchments within North-East England



**Table 1.1 A summary of benchmarking catchments in North-East England**

Catchment	Name	Benchmark?	Rainfall (average annual; mm)	Built up %	Chalk Stream?	BFI	Reservoir in catchment?	Catchment Area (km ²)
23007	Derwent at Rowlands Gill	N	849	4.22%	N	0.194–0.388	Y	243.99
24003	Wear at Stanhope	N	1279	0.34%	N	0.194–0.388	Y	173.49
25020	Skerne at Preston le Skerne	N	654	10.55%	N	0.388–0.455	N	152.76
23006	South Tyne at Featherstone	Y	1331	0.29%	N	0.194–0.388	N	323.05
25006	Greta at Rutherford Bridge	N	1127	0.14%	N	0.194–0.388	N	86.81
25012	Harwood Beck at Harwood	N	1577	0%	N	0.194–0.388	N	24.58

As shown in Table 1.1, the selected catchments have a relatively wide range of annual average rainfall, built-up area and total area. Unfortunately, the base flow index (BFI) has a narrow range in North-East England and is relatively low. The BFI is an indicator of the contribution of groundwater to river base flow and is therefore of relevance to flow. In addition, there are no chalk streams within North-East England, so this selection of benchmarking catchments does not allow the relative performances of the models on chalk catchments to be assessed. One National River Flow Archive (NRFA) 'benchmark' catchment is included in the selected benchmark catchments. The NRFA definition of 'benchmark' is not to be confused with the benchmarking reported on in this document. A NRFA benchmark catchment is a catchment with minimal anthropogenic influences on river flows; therefore, flows from these catchments can be regarded as naturalised flows. More information on benchmark catchments is given by UKCEH.

If timely access to QUBE licencing nationally was not a restriction, a more geographically dispersed selection of benchmark catchments can be recommended (Figure 1.2).

Figure 1.2 **Positions of recommended benchmark catchments for the RDMAI Open Flow Model if obtaining QUBE licences are not a restriction**



These recommended catchments are described in more detail in Table 1.2.

**Table 1.2 A summary of recommended benchmark catchments for the RDMAI Open Flow Model if QUBE licences are not a restriction**

Catchment	Name	Region	Benchmark?	Rainfall (average annual; mm)	Built up %	Chalk Stream?	BFI	Catchment Area (km ²)
24003	Wear at Stanhope	North East	N	1279	0.34%	N	0.194–0.388	173.49
25020	Skerne at Preston le Skerne	North East	N	654	10.55%	N	0.388–0.455	152.76
23007	Derwent at Rowlands Gill	North East	N	849	4.22%	N	0.194–0.388	243.99
73010	Leven at Newby Bridge	North West	N	2172	2.75%	N	0.388–0.455	247.97
27009	Ouse at Skelton	Yorkshire and The Humber	N	899	3.38%	N	0.388–0.455	3300.8
30012	Stainfield Beck at Cream Poke Farm	East Midlands	Y	631	1.64%	Y	0.636–0.955	38.22
28080	Tame at Lea Marston Lakes	West Midlands	N	716	56.75%	N	0.455–0.527	795.1



Catchment	Name	Region	Benchmark?	Rainfall (average annual; mm)	Built up %	Chalk Stream?	BFI	Catchment Area (km ²)
34007	Dove at Oakley Park	East of England	N	585	4.13%	N	0.388–0.455	139.37
36010	Bumpstead Brook at Broad Green	East of England	Y	588	3.51%	N	0.388–0.455	27.58
33014	Lark at Temple	East of England	N	593	8.11%	Y	0.636–0.955	274.04
33053	Granta at Stapleford	East of England	N	576	5.1%	Y	0.636–0.955	113.98
33021	Rhee at Burnt Mill	East of England	N	559	5.62%	Y	0.636–0.955	308.05
41003	Cuckmere at Sherman Bridge	South East	N	814	6.07%	N	0.388–0.455	130.33
46003	Dart at Austins Bridge	South West	Y	1771	1.75%	N	0.388–0.455	249.99



1.3 Benchmarking Methodology

Daily average timeseries QUBE flows (m³/s) for the chosen benchmark catchments (Table 1.1; Figure 1.1) were extracted from the [Online QUBE portal](#). Importantly, when QUBE flows were extracted, under advanced settings, the choice to inform QUBE flows by nearby flow gauges were de-selected. This forces QUBE estimations of flow to be completely simulation based, allowing for a fairer comparison to the RDMAI Open Flow Model. In this way, the flows provided by QUBE are based purely on the rainfall-runoff model and are not informed by nearby flow gauge measures.

RDMAI Open Flow Model simulated daily average flows (m³/s) for the selected catchment, along with gauged flows, were obtained from the model validation dataset in a spreadsheet format. The QUBE flows were added to same spreadsheet, ensuring temporal matching between the flow timeseries.

A series of Python scripts were generated to read in and to construct comparative plots of the flows in these collated datasets (gauged flows, RDMAI Open Flow Model simulated flows and QUBE simulated flows). The scripts generate timeseries and flow duration curve graph comparisons between the flow datasets. In addition, goodness-of-fit statistics that compare the flow simulation for each model to the gauged flow are generated.

Nash-Sutcliffe Efficiency (NSE)

Despite reported inadequacies with this metric, it is widely used in hydrological studies to assess the predictive power of hydrological models. Therefore, it is included within this report, as many practitioners in the water industry are familiar and comfortable with this metric. The NSE can give a value between 1 and -infinity, with 1 indicating a perfect fit. The NSE is calculated as:

$$NSE = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (1)$$

Where O_i is observed discharge, P_i is predicted discharge, and $\bar{O}$ is the mean of observed discharge.

Percent Bias (PBIAS)

Percent Bias (PBIAS) measures the average negative deviation of the predicted data from the observed data. A PBIAS of 0% is the optimum value. A negative and positive PBIAS indicates an overestimation and underestimation by the model, respectively. It is calculated as:

$$PBIAS = \frac{\sum_{i=1}^n (O_i - P_i)}{\sum_{i=1}^n O_i} \times 100 \quad (2)$$

Root Mean Square Error (RMSE)



The Root Mean Square Error (RMSE) is the average difference between model predicted and observed. The lower the difference, the better the model fits to observed data. It is calculated as:

$$RMSE = \sqrt{\sum (P_i - O_i)^2 / n} \quad (3)$$

Where n is the number of individual comparisons.

1.4 Benchmarking Limitations

As discussed previously, due to the restrictions on available QUBE data, a limited set of benchmark catchments were available for comparison. Benchmarking catchments were restricted to North-East England – the operational area of Northumbrian Water Limited.

Ideally, human influenced QUBE flows should be used in the benchmarking. The QUBE documentation indicates that these flows can be generated. However, when extracting the flows, the QUBE interface indicated that extraction of human influenced flows is restricted to the regulator, due to confidential data being used to estimate these flows. Therefore, this benchmarking uses naturalised QUBE flows. This limitation should be considered when interpreting the results of the benchmarking.

There were missing data in the observed flows for the benchmark catchments. RDMAI Open Flow Model estimates of flow from the validation did not include flows for these missing periods. However, this does not change the results of the benchmarking: comparison of model estimates of flow with observed flow can only be done for existing gauged flow data.



2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
Installation & Run Process	The QUBE flows are pre-generated by a rainfall-runoff model and then made available to download from a website. Therefore, there is no need for the user to install or run a model.	The model can be cloned from GitHub. Installing the model dependencies can be achieved through a single command line call. A sequence of numbered scripts, in the form of Jupyter Notebooks, facilitates the download and preparation of data and the running of the trained model for a prepared example. All the above steps take 1 -2 hours to complete. Training the model would require the steps mentioned above plus the time taken to train the model and was not assessed in this report. Due to licensing issues, the flow model would need to be trained by each user, before use.



Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
Data inputs	QUBE flows can be accessed from a website . Since the user does not need to run the model, no data inputs need be considered.	The RDMAI Open Flow Model requires the CAMELS-GB dataset as input. This is downloaded and processed automatically through a script. If the user wanted to run the RDMAI Open Flow Model in a non-CAMELS-GB defined catchment, significant pre-processing would be needed to create the relevant data inputs.
Licensing Requirements	QUBE flows require a paid-for licence.	The RDMAI Flow Model is open source and free, and all input data are openly available.



2.2 Model Outputs

Table 2.2 **Comparison of model outputs**

Model Outputs	QUBE (Existing Industry Model)	RDMAI Model
Data Format	QUBE flows can be downloaded as a comma-separated file (CSV).	Outputs of the RDMAI Open Flow Model are flexible. The flow outputs can be generated as a comma-separated file (CSV), while the provided scripts also allow various graphing and mapping outputs.
Temporal Resolution	QUBE flow outputs are daily mean flow rate in m ³ /s.	RDMAI Open Flow Model outputs are daily mean flow rate in m ³ /s.
Geographic Constraints	United Kingdom and Ireland.	England.

2.3 Time and Cost Comparison

Apart from the initial time taken to install the RDMAI Open Flow Model and input data, which takes around 2 hours, running the trained model takes minutes. Training the model would require additional time, which was not assessed in this report. Running a traditional hydrological model is likely to take considerably longer, considering the collection of data required, the setup of the model, and model calibration and validation. Extraction of QUBE flow data takes around 30 minutes per site, but this is a paid-for product. It should be noted that at present, QUBE is only available for lease on a regional basis, aimed at use by the UK regulators and water companies. These leases may be negotiated on an individual basis.

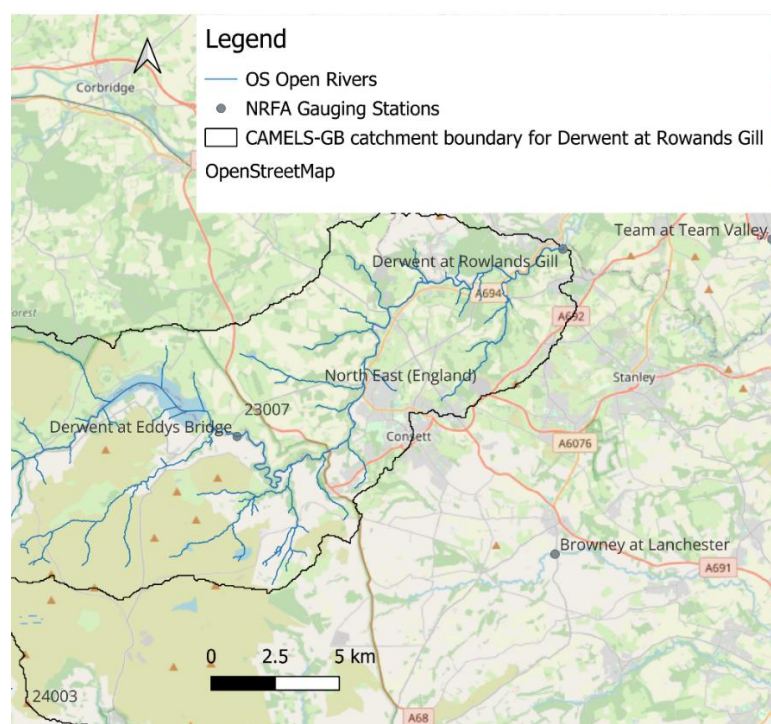
The Water Evaluation and Planning ([WEAP](#)) model has built-in rainfall-runoff hydrology models and can provide estimates of human-impacted river flow. A multi-user corporate licence for this software costs \$10,000/year.

2.4 Model Performance

2.4.1 Derwent at Rowlands Gill

As shown in Figure 2.1, the catchment contributing flow to Derwent at Rowlands Gill contains a relatively large reservoir – Derwent Reservoir. As shown in Table 1.1, the catchment does contain some built-up land (4.22%). The contributing catchment is one of the bigger catchments assessed for the benchmarking, at around 244 km².

Figure 2.1 Map of the catchment contributing flow to Derwent at Rowlands Gill





As shown in Figure 2.2, Figure 2.3 and Figure 2.4, QUBE and RDMAI Open Flow estimated flows for Derwent at Rowlands Gill were of relatively poor accuracy when compared to gauged flows. This is indicated by NSEs of -0.46 and -0.73 , respectively, shown in Figure 2.2. Figure 2.3 shows the timeseries differences between estimated and measured flows for one year; whilst it is clear that the estimated flows capture the temporal changes in measured flows, the estimated flows are consistently too high. The FDCs in Figure 2.4 shows that both QUBE and the RDMAI Open Flow model tend to over-estimate intermediate flows in this catchment.

Figure 2.2 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Derwent at Rowlands Gill

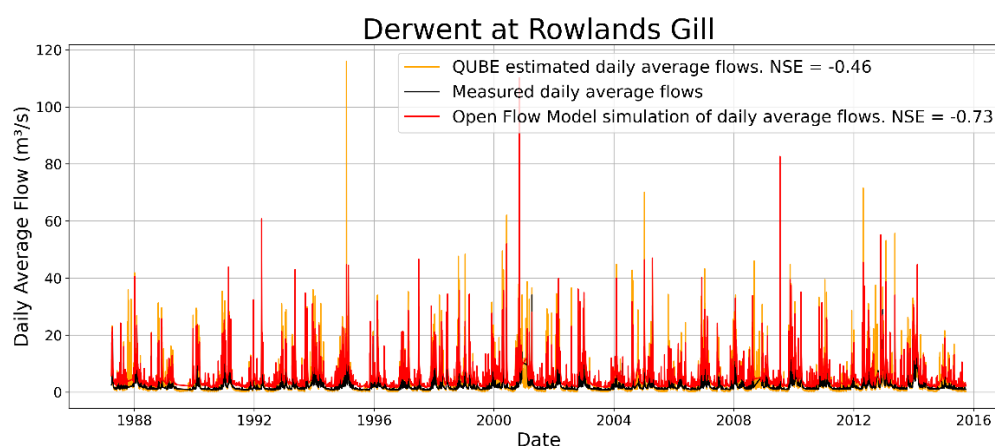


Figure 2.3 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Derwent at Rowlands Gill focusing on one year - 1993

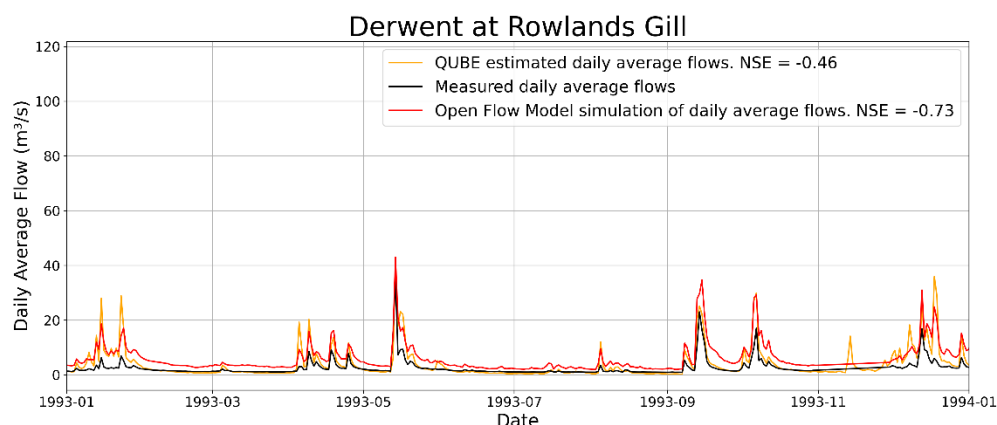
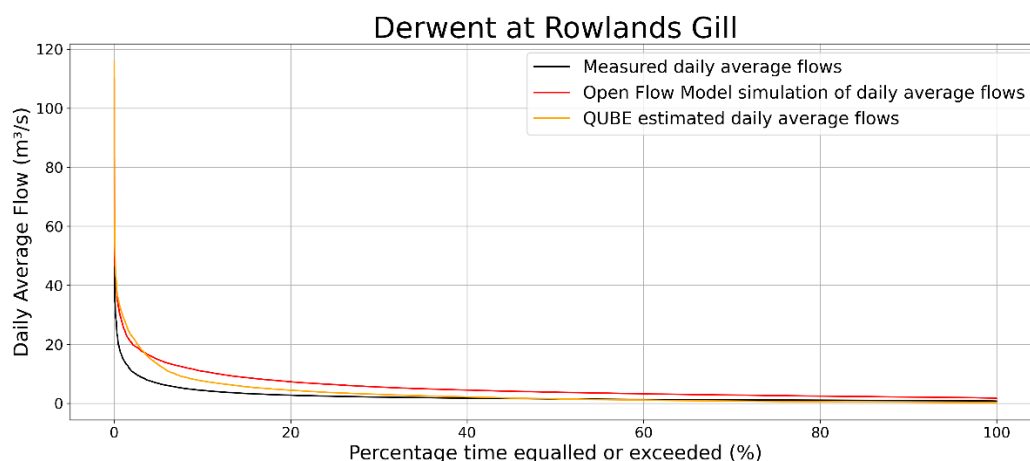


Figure 2.4 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Derwent at Rowlands Gill

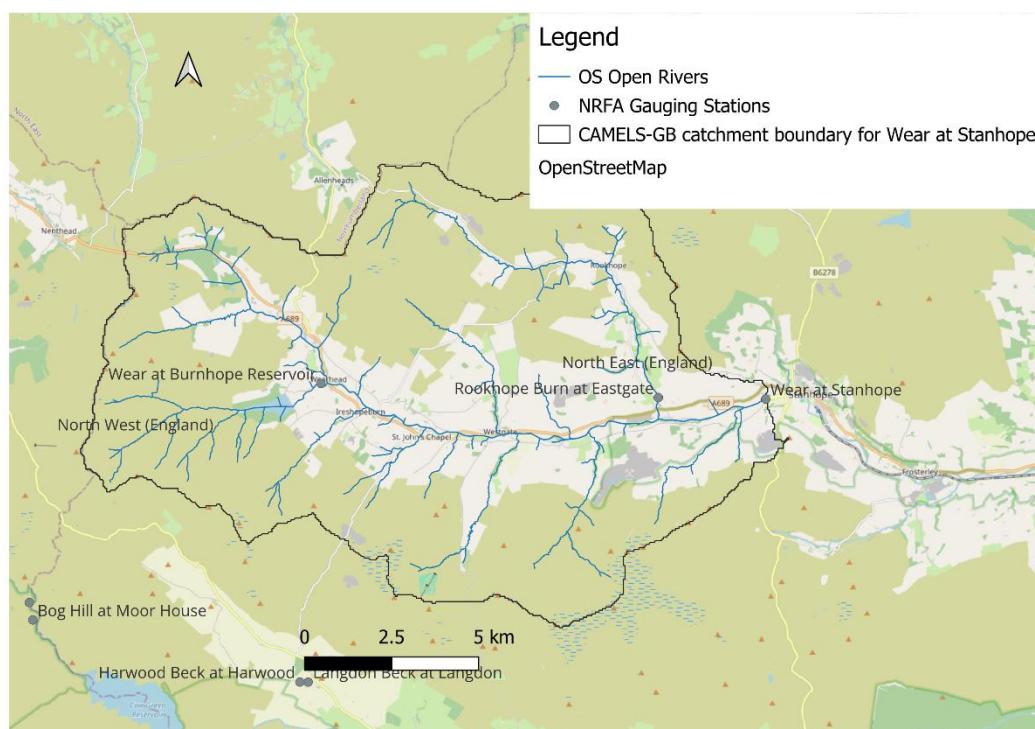


An examination of the catchment contributing to Derwent at Rowlands Gill can explain the reasons for the differences between QUBE, the RDMAI Open Flow Model estimated flows and gauged flows. As shown in Figure 2.1, Derwent Reservoir is upstream of the flow gauging point Derwent at Rowlands Gill. This reservoir would alter the natural flows of the river, likely explaining the higher frequency of moderate flows estimated by the two models, shown in Figure 2.4. Since the QUBE flows extracted are natural flows, the effect of the reservoir would not be evident in the extracted flows.

2.4.2 Wear at Stanhope

Figure 2.5 shows that the catchment contributing flow to Wear at Stanhope contains a reservoir. The catchment contains relatively little built up area (0.34%) and is of intermediate size in relation to the benchmarked catchments (around 173 km²).

Figure 2.5 Map of the catchment contributing flow to Wear at Stanhope



As shown in Figure 2.6 and Figure 2.7, the QUBE model estimated flow was a good fit to gauged flow at Wear at Stanhope, with an NSE of 0.92. The flows provided by the RDMAI Open Flow Model, while not being as good as QUBE, were also representative of gauged flows, with an NSE of 0.84. An examination of the FDCs in Figure 2.8 indicate that both the QUBE and RDMAI Open Flow model provide flows with a similar frequency to gauged flows. The QUBE flows slightly over-estimate gauged flows, which is understandable given that the QUBE flows are naturalised.

Figure 2.6 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Wear at Stanhope

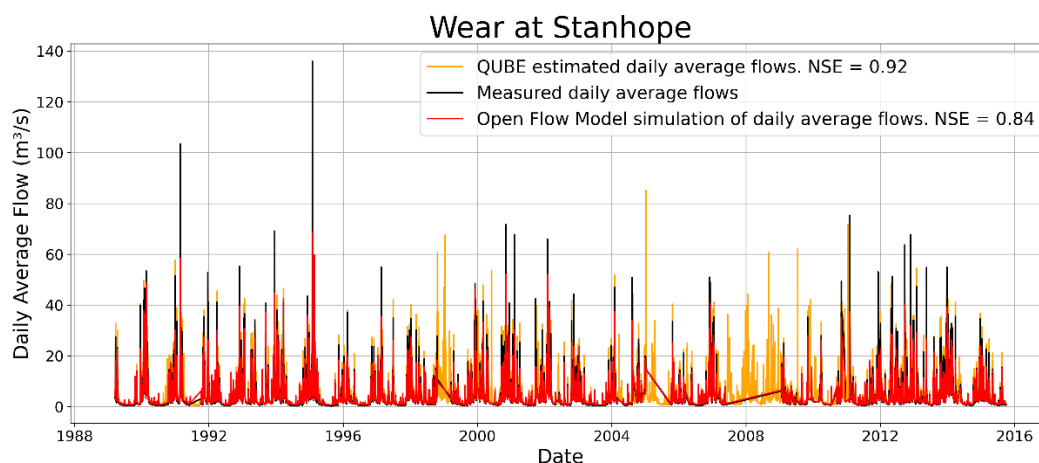
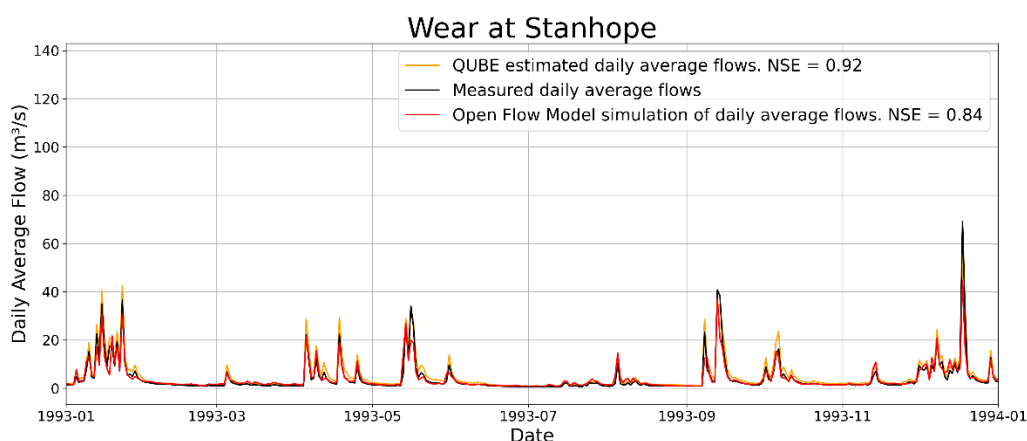
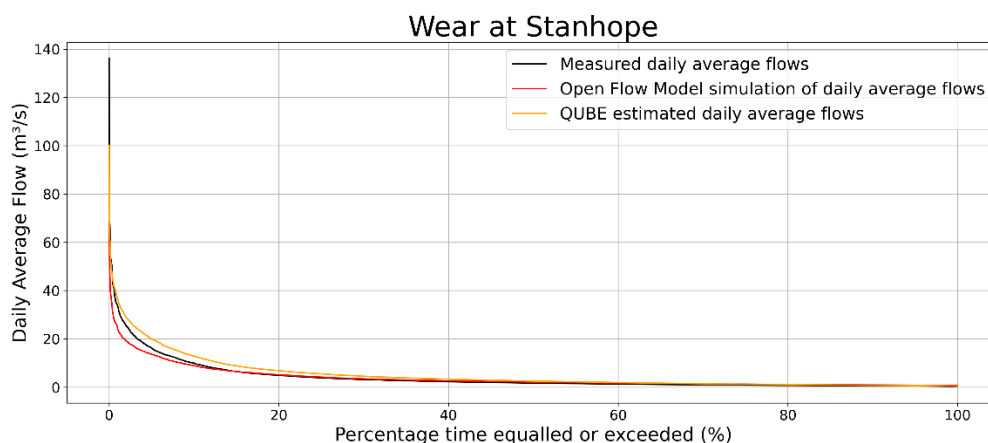


Figure 2.7 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Wear at Stanhope focusing on one year - 1993



As shown in Figure 2.5, Burnhope Reservoir is in the upper catchment contributing to Wear at Stanhope. This is a relatively small reservoir, but it might in part explain why the QUBE model flow estimations are higher than the observed. In theory, the RDMAI Open Flow Model should consider the effects of reservoirs in the catchment on estimated flow, and for this catchment, the model seems to have taken sufficient account of the reservoir in its flow estimations.

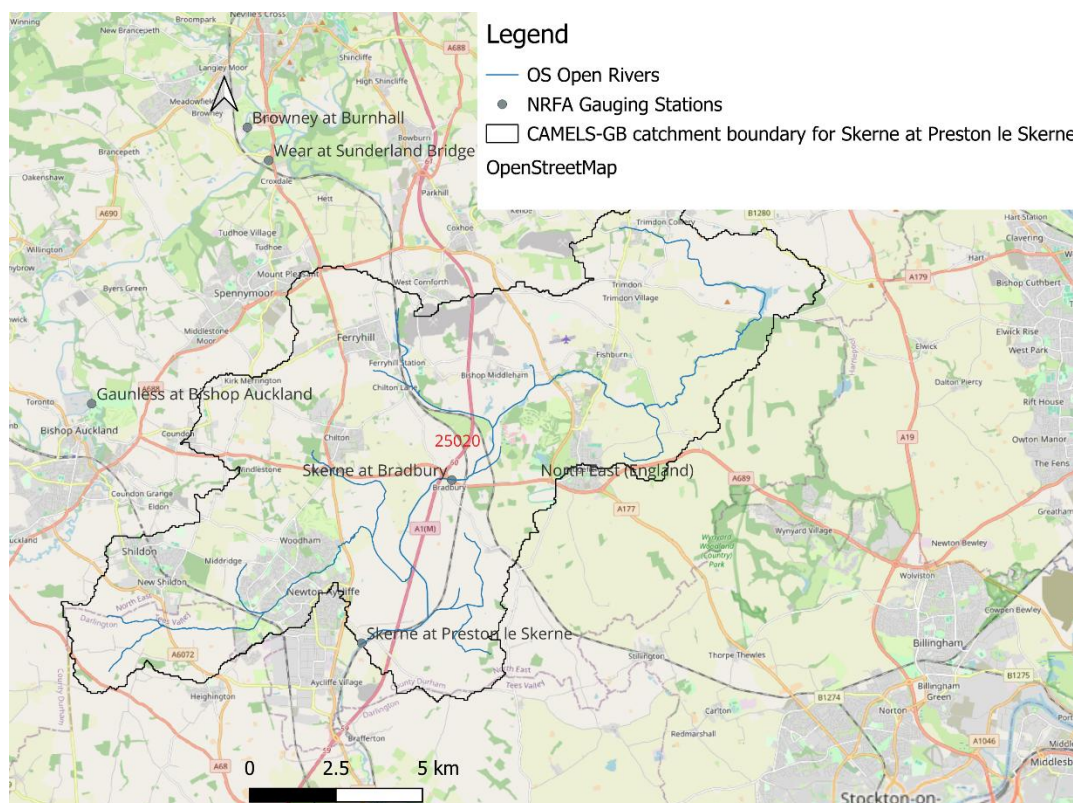
Figure 2.8 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Wear at Stanhope



2.4.3 Skerne at Preston le Skerne

The catchment contributing flow to Skerne at Preston le Skerne (Figure 2.9) contains no major reservoirs. Table 1.1 shows that this catchment contains a relatively high proportion of built-up area (10.55%). The catchment has the lowest rainfall amongst the benchmarked catchments (654 mm/year) and is of average size (around 153 km²).

Figure 2.9 Map of the catchment contributing flow to Skerne at Preston le Skerne



As shown in Figure 2.10 and Figure 2.11, QUBE gives estimated flows that are relatively representative of gauged flows, with an NSE of 0.94. The RDMAI model has a lower NSE of 0.43. Focusing on one year of timeseries difference, Figure 2.11 shows that while the RDMAI model captures temporal changes in measured flow, the estimated flows are consistently too low. Figure 2.12 shows that QUBE flows represent the gauged frequency of flows well, whereas the RDMAI Open Flow Model underestimates gauged flows.

Figure 2.10 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Skerne at Preston le Skerne

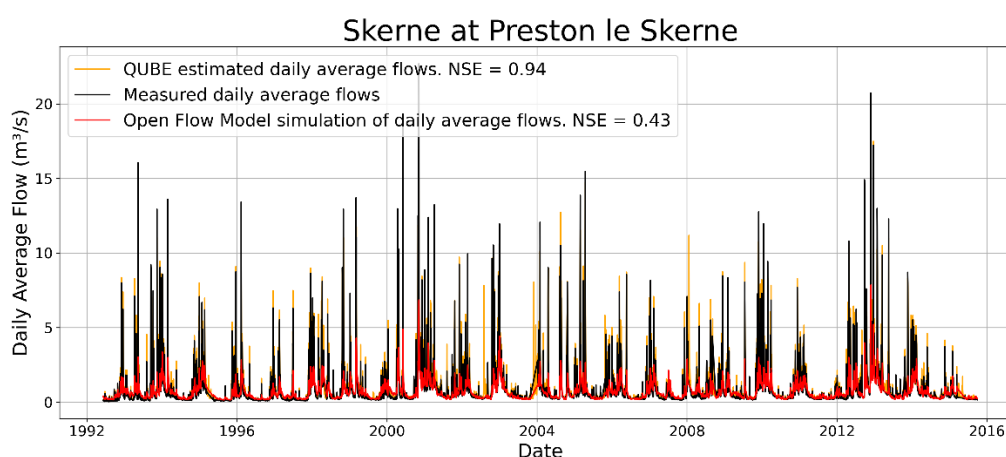


Figure 2.11 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Skerne at Preston le Skerne focusing on one year - 1993

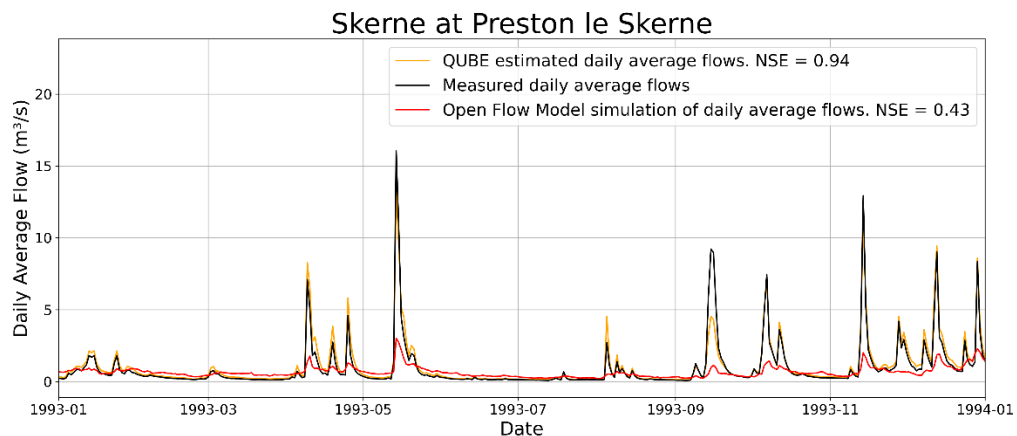
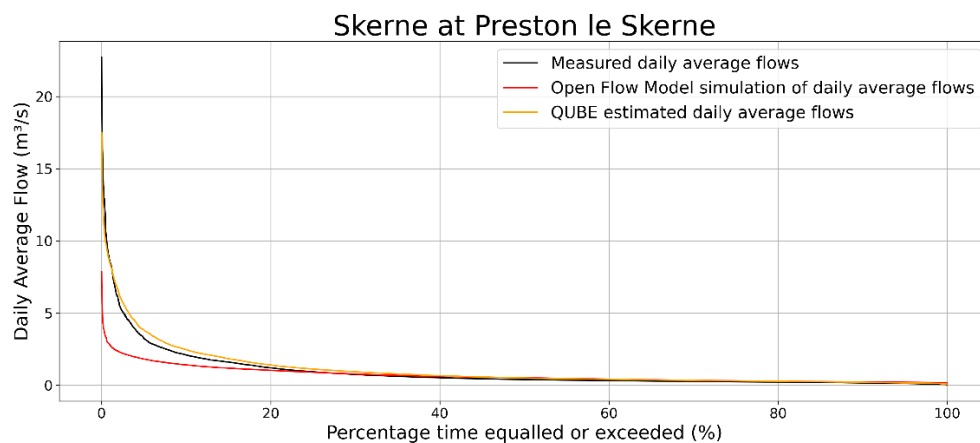


Figure 2.12 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Skerne at Preston le Skerne



As shown in Figure 2.9, there are no major reservoirs in the catchment contributing to flow at Skerne at Preston le Skerne. However, this catchment has the largest proportion of built-up area among the benchmark catchments, at around 10.6% (Table 1.1), which may be contributing to the model performances. This is because it is reasonable to correlate built-up area with human influences on flow that the RDMAI model may struggle to represent, such as water abstractions.

2.4.4 South Tyne at Featherstone

As shown in Figure 2.13, the catchment contributing to flow at South Tyne at Featherstone contains no major reservoirs. Table 1.1 also indicates the catchment to be 'un-impacted', with negligible built-up area. This catchment is also included in the NRFA list of UK benchmark

network (UKBN) catchments, i.e., a catchment relatively unimpacted by human influences affecting flow. Therefore, gauged flows from this catchment can be regarded as naturalised flows. The catchment area is the largest amongst the catchments assessed (approximately 323 km²).

Figure 2.13 Map of the catchment contributing flow to South Tyne at Featherstone



Figure 2.14 and Figure 2.15 show that both the QUBE and RDMAI Open Flow Model estimated flows were representative of gauged flows, with NSEs of 0.96 and 0.89, respectively.

Figure 2.16 shows that both models also gave good representation of the observed frequency in flow.

Figure 2.14 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at South Tyne at Featherstone

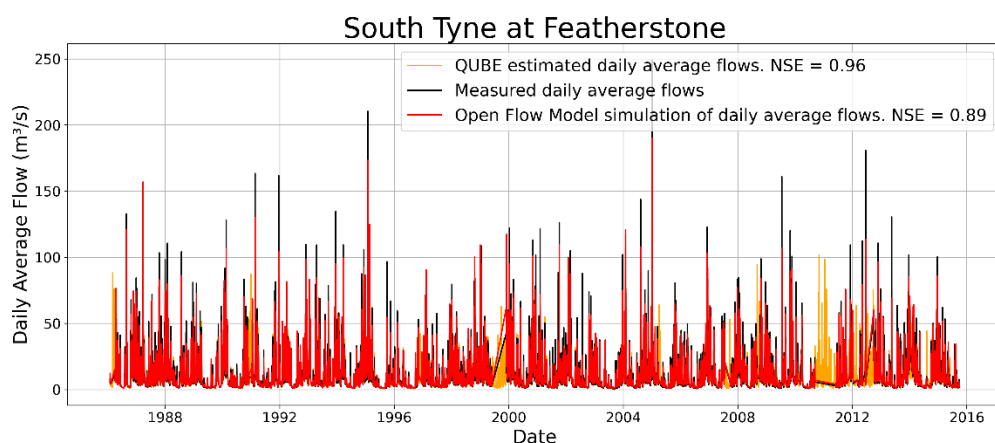


Figure 2.15 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at South Tyne at Featherstone focusing on one year - 1993

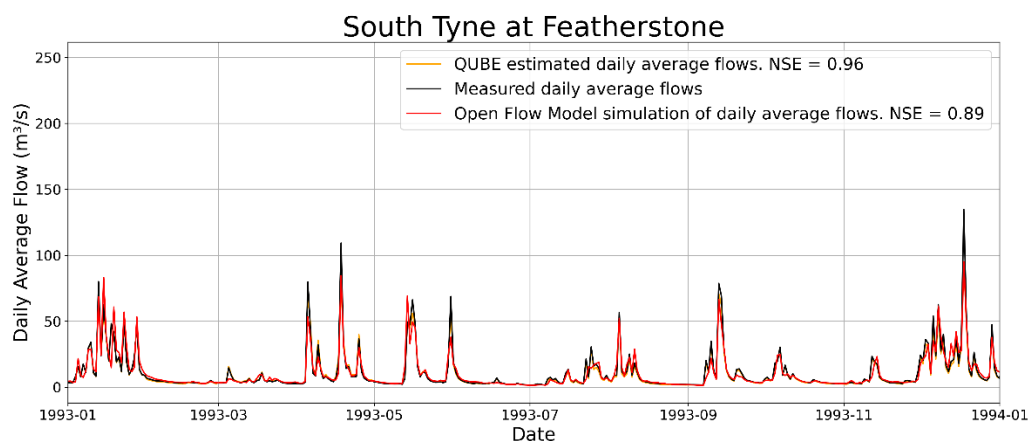
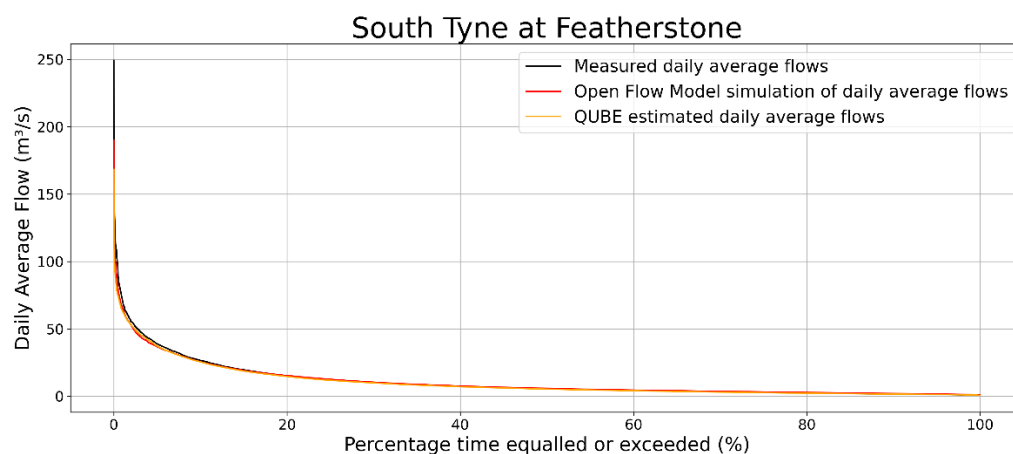


Figure 2.16 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at South Tyne at Featherstone



2.4.5 Greta at Rutherford Bridge

As shown in Figure 2.17, there are no major reservoirs or lakes in the catchment contributing flow to Greta at Rutherford Bridge. Table 1.1 also shows that this catchment is un-impacted by human influence, with a negligible proportion of built-up area (0.14%). The catchment area is relatively small at around 87 km².

Figure 2.17 Map of the catchment contributing flow to Greta at Rutherford Bridge

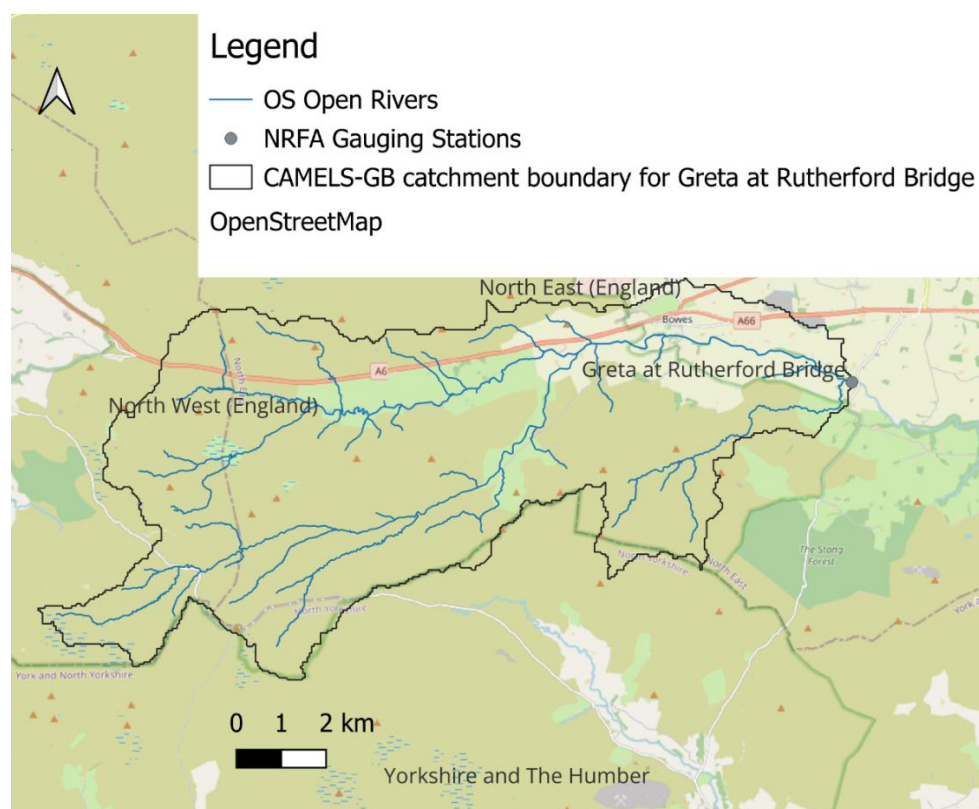




Figure 2.18 and Figure 2.19 indicate that QUBE estimated flows are representative of gauged flows, with an NSE of 0.93. This is not the case for RDMAI Open Flow Model estimated flows, with a NSE of -0.22. These model performances are reflected in Figure 2.20, with the QUBE flows being relatively representative of the frequency in gauged flows but flows by the RDMAI Open Flow Model considerably underestimating higher gauged flows. As shown in Figure 2.23, RDMAI Open Flow Model estimated flows remain relatively unresponsive, even when there are major spikes in measured flows.

Figure 2.18 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Greta at Rutherford Bridge

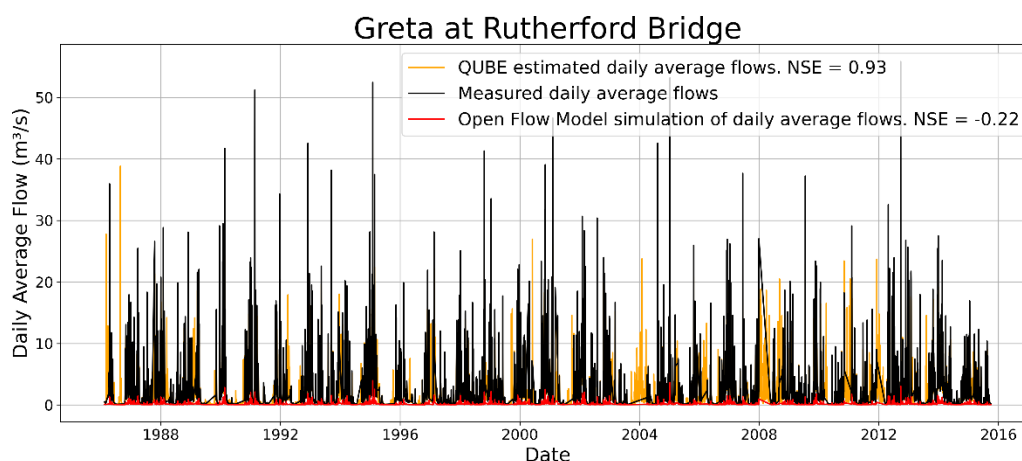


Figure 2.19 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Greta at Rutherford Bridge focusing on one year - 1993

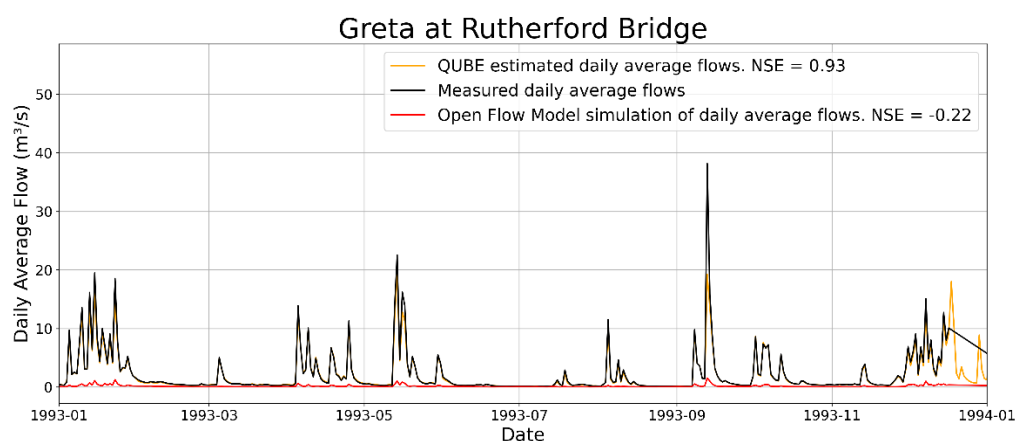
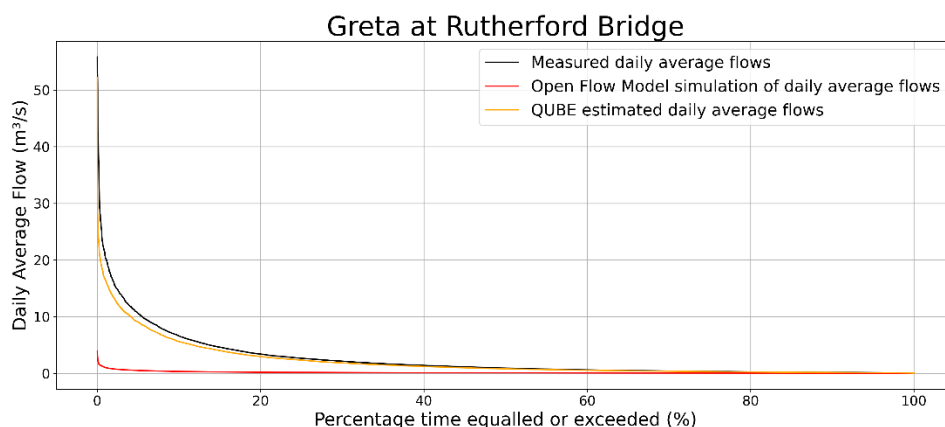


Figure 2.20 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Greta at Rutherford Bridge



2.4.6 Harwood Beck at Harwood

Figure 2.21 shows no major reservoirs in the upstream catchment contributing flow to Harwood Beck at Harwood. Table 1.1 also indicates a relatively unimpacted and small catchment, with a built-up area proportion of only 0.14% and catchment area of around 25 km².

Figure 2.21 Map of the catchment contributing flow to Harwood Beck at Harwood

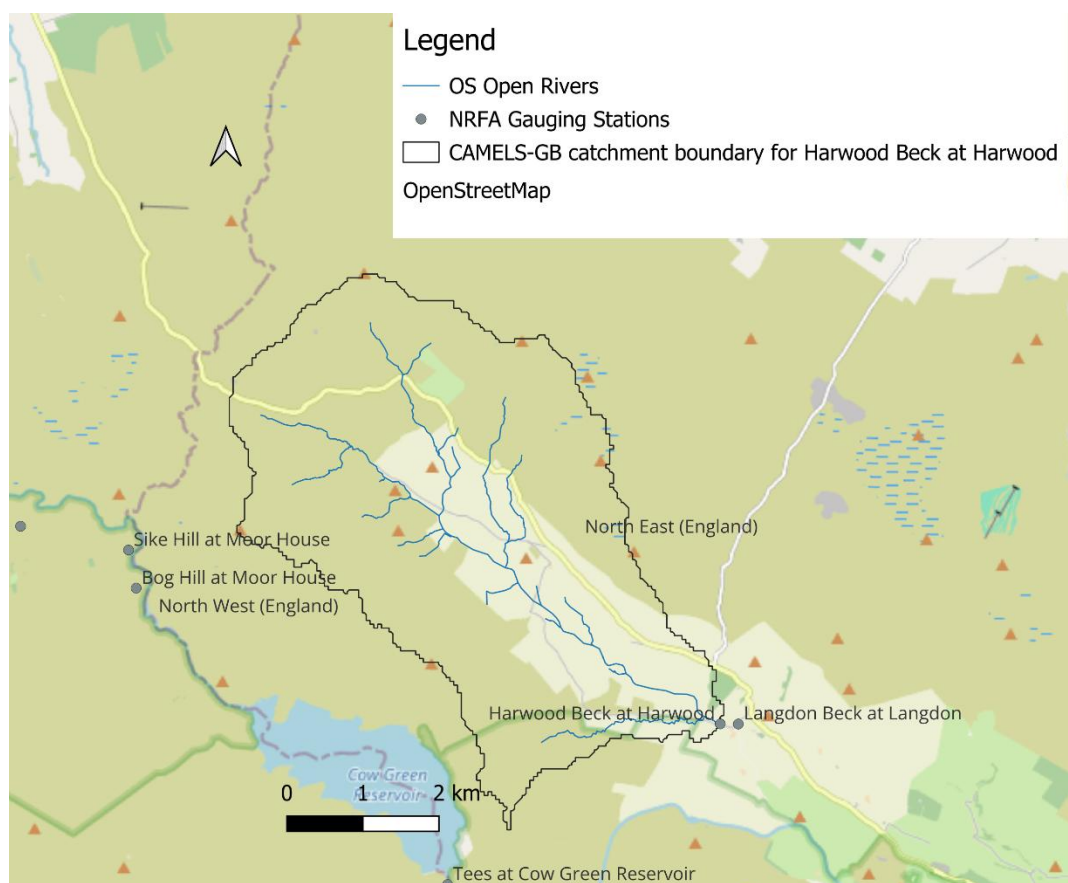




Figure 2.22 and Figure 2.23 show that QUBE flows are representative of gauged flows, with an NSE of 0.95. The RDMAI Open Flow Model flows are significantly less representative, with an NSE of 0.05. These differences are also reflected in Figure 2.23 with QUBE flows representing the frequency of gauged flows well, whilst the RDMAI Open Flow Model substantially underestimates the higher gauged flows. As shown in Figure 2.23, the flows estimated by the RDMAI Open Flow Model respond to changes in measured flow, but consistently under simulate flow.

Figure 2.22 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Harwood beck at Harwood

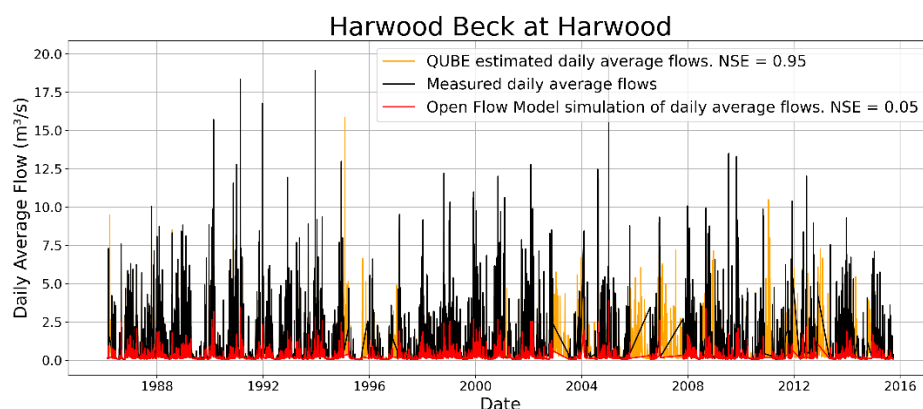


Figure 2.23 Timeseries comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Harwood beck at Harwood focusing on one year - 1993

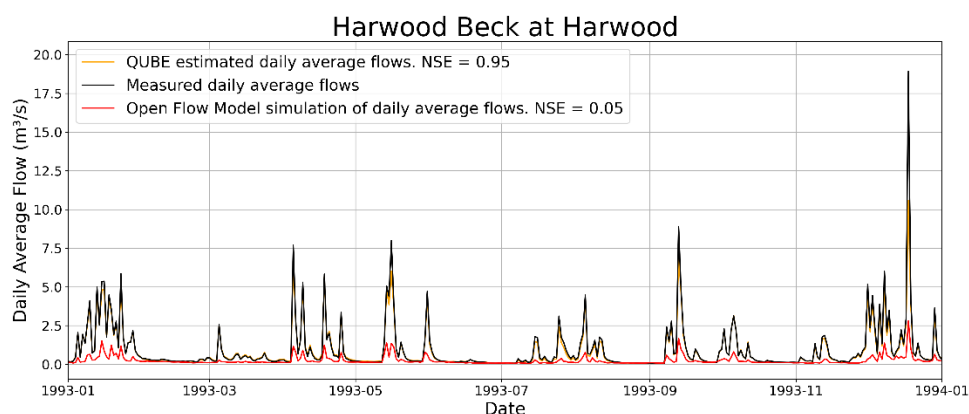
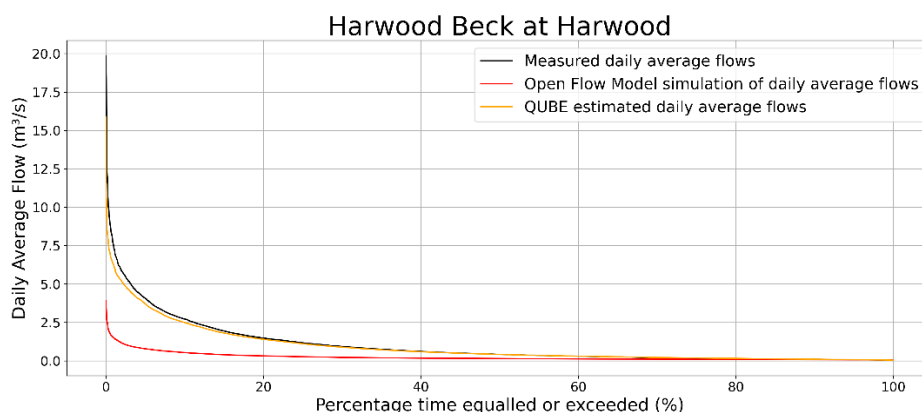


Figure 2.24 Flow duration curve comparison between gauged flows, RDMAI Open Model simulated flows and QUBE simulated flows at Harwood beck at Harwood



2.5 Model Performance Summary Statistics

Table 2.3 shows a summary of the performances of the QUBE and RDMAI Open Flow Model estimated flows when compared to observed flows.

- Table 2.3 shows a relatively poor performance by QUBE for **Derwent at Rowlands Gill**. The NSE and PBIAS, as well as the FDCs shown in Figure 2.4, confirm that the QUBE model over-estimates observed flow. The RDMAI Open Flow Model also over-estimates flow for this catchment. As shown in Figure 2.1, the contributing catchment contains Derwent Reservoir. Since QUBE flows are naturalised flows, they do not reflect the influence of the reservoir, which could explain the differences to observed flows. The RDMAI Open Flow Model in theory should account for human influences. However, for this catchment, the model does not appear to fully account for the influence of Derwent Reservoir.
- The NSE values for **Wear at Stanhope** confirm that both models give a relatively good approximation of the temporal pattern in observed flows. Both models also give a relatively good representation of gauged flow frequency. QUBE tends to over-estimate flow, as indicated by the PBIAS and the FDC shown in Figure 2.8. In contrast, the RDMAI Open Flow Model provides a more accurate representation of flow frequency. The contributing catchment contains Burnhope Reservoir, which may be contributing to the over-estimation of flow by QUBE (see the negative PBIAS in Table 2.3), which is understandable, as QUBE gives natural flows, without the influence of the reservoir. The RDMAI Open Flow Model appears to account for the influence of the reservoir in this catchment.
- Both models give relatively good NSEs for **Skerne at Preston le Skerne**. The PBIAS values shown in Table 2.3 indicate that the QUBE flows slightly over-estimate observed, while the RDMAI Open Flow Model flows slightly underestimate observed. These PBIAS values are re-enforced by the FDCs in Figure 2.12. Figure 2.9 shows no



obvious impoundments that could be influencing the relative performances of the two models.

- Table 2.3 shows that both models perform relatively well in representing both the temporal pattern and frequency in flow magnitudes at **South Tyne at Featherstone** (Figure 2.14, Figure 2.15, Figure 2.15,). As shown in Figure 2.13, there are no major reservoirs in the contributing catchment that could be influencing flows.
- The RDMAI Open Flow Model estimated flows are a relatively poor match to observed flows at **Greta at Rutherford Bridge**. The low NSE and high positive PBIAS in Table 2.3 indicate that the RDMAI Open Flow Model considerably under-estimates gauged flows. This observation is re-enforced by Figure 2.18, Figure 2.19 and Figure 2.20. In contrast, QUBE estimated flows are generally a good match to observed. It is not clear what might be contributing to the mismatch between RDMAI Open Flow Model flows and gauged flows. Looking at Figure 2.17, there are no obvious human influences, such as reservoirs, that could be affecting flow. Among the benchmarked catchments, this catchment has a relatively small area, at around 87 km². It is possible that the RDMAI Open Flow Model struggles to accurately represent flow in smaller catchments.
- Table 2.3 shows that the QUBE model estimated flows for **Harwood Beck at Harwood** are generally a good match to observed flows, whereas the NSE and PBIAS indicate that the RDMAI Open Flow Model considerably under-estimates gauged flows. This is further confirmed by Figure 2.22, Figure 2.23 and Figure 2.23. Focusing on one year of timeseries flow, Figure 2.23 shows that the flow estimates by the RDMAI Open Flow Model do respond to temporal spikes in measured flow, but consistently under-estimate flow. Looking at Figure 2.21, there are no obvious human influences, such as reservoirs that could be affecting flow. However, again, this is a relatively small catchment, with an area of 25 km², indicating a possible shortcoming to the RDMAI Open Flow Model in representing smaller catchments.

In general, the QUBE model provided flow estimates that were a good match to gauged flows. In some cases, where the QUBE model did provide flow estimates that were clearly higher than gauged flows, such as for Derwent at Rowlands Gill and Wear at Stanhope, there are reservoirs in the upstream contributing catchments. Since the available QUBE flows are naturalised flows, we would not expect these flows to reflect human influences on flow, such as reservoir flow regulation.

The performances of the RDMAI Open Flow Model are more difficult to explain. The model appeared to over-estimate flow at Derwent at Rowlands Gill, possibly not accounting sufficiently for the effect of the reservoir. However, it did sufficiently account for the smaller reservoir affecting flow at Wear at Stanhope. In addition, it considerably under-estimated flows for some catchments with no clear human influences on flow, such as at Greta at Rutherford Bridge and at Harwood Beck at Harwood. However, these two catchments have relatively small catchment areas, indicating a possibly shortcoming in the RDMAI Open Flow Model in estimating flow for smaller contributing catchments.



Abstractions are another human influence, besides reservoirs, that could be affecting flows. Unfortunately, the positions of abstractions are not openly available, so this has not been investigated as part of this study.

The RDMAI Open Flow Model relies primarily on the CAMELS-GB dataset to inform estimated river flow. It is recommended that future studies using the RDMAI Open Flow Model interrogated and cross-check the CAMELS-GB dataset with other information for a subset of good and poor performing catchments, according to RDMAI Open Flow Model flows. This is to determine if there are any inconsistencies or missing information between the CAMELS-GB dataset and other data, such as the presence of reservoirs and reservoir capacity. In this way, additional input datasets to the RDMAI Open Flow Model can be identified that could possibly increase the accuracy of the model.



Table 2.3 **Summary of performance metrics for the comparison of QUBE and RDMAI Open Flow Model estimated flows with gauged flows**

Gauge ID	Gauge Name	Observed	QUBE			RDMAI Open Flow Model		
		Mean daily flow (m ³ /s)	NSE	RMSE (m ³ /s)	PBIAS (%)	NSE	RMSE (m ³ /s)	PBIAS (%)
23007	Derwent at Rowlands Gill	2.38	-0.46	3.86	-41.01%	-0.73	4.20	-130.55%
24003	Wear at Stanhope	3.99	0.92	1.85	-28.88%	0.84	2.69	3.34%
25020	Skerne at Preston le Skerne	0.92	0.94	0.37	-15.93%	0.43	1.20	25.09%
23006	South Tyne at Featherstone	10.78	0.96	3.04	5.51%	0.89	5.02	1.02%
25006	Greta at Rutherford Bridge	2.50	0.93	1.12	10.44%	-0.22	4.70	93.63%
25012	Harwood Beck at Harwood	1.01	0.95	0.33	8.01%	0.05	1.49	75.72%



3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found in the summary, at the start of the report.

3.2 Conclusion

This benchmarking faced various limitations. The most important were:

- Only a limited number of benchmarking catchments could be examined because of limitations to available QUBE flow licences.
- The available QUBE flows provide naturalised flows, which do not reflect human influences on flow, such as from reservoirs, abstractions and discharges.

The QUBE flows generally were a good match to gauged flows for the benchmark catchments. The instances where QUBE flows notably overestimated gauged flows could be explained by the presence of reservoirs in the upstream catchment.

In one case, the RDMAI Open Flow Model did give estimates of flow reflecting the presence of an upstream reservoir (Burnhope Reservoir). However, the model seemed to under-estimate the influence of the larger Derwent Reservoir by overestimating flow. For two relatively small benchmark catchments, despite there being no reservoirs in the contributing catchments, the RDMAI Open Flow Model considerably under-estimated gauged flow.

Out of the six benchmark catchments, the RDMAI Open Flow Model gave reasonable estimates for 3 (good match to gauged flow temporal pattern), and good estimates for 2 (good match to both gauged flow temporal pattern and flow magnitude frequency). The RDMAI Open Flow Model is open source and freely available, which is a considerable benefit, highlighted by the challenges encountered here in trying to secure QUBE flows. In addition, available QUBE flows do not reflect human influences, whereas, the RDMAI Open Flow Model has the potential to represent these influences in estimated flows. Therefore, there is future potential for the application of the RDMAI Open Flow Model for wastewater management applications, water resource planning, and flow risk modelling subject to further refinement and testing.

Suggested further improvements can be an examination of the CAMELS-GB dataset against other available datasets, to identify any inconsistencies or additional data that can be input into the RDMAI Open Flow Model. The results of the benchmarking indicate a possible shortcoming in the representation of the influence of reservoirs on estimated flows. Due to the limited number of benchmark catchments assessed, other possible input data shortcomings are difficult to identify. Abstractions are possibly a human influence not sufficiently accounted for by the RDMAI Open Flow Model, particularly given the sensitivity surrounding locations of water abstractions and subsequent lack of open data for abstractions. In particular, the North East of



England has a lot of groundwater abstractions, and the RDMAI Open Flow Model does not consider the influences of these abstractions on river flow due to surface water-groundwater interactions. The influences of surface water and groundwater abstractions on river flow may represent an interesting area for further investigation.



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